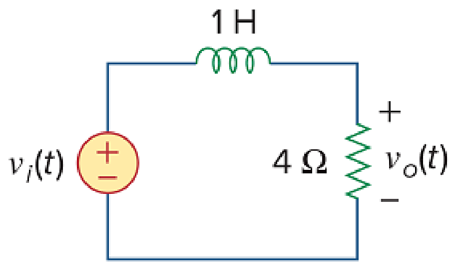
Chapter 18, Problem 7PP

Problem

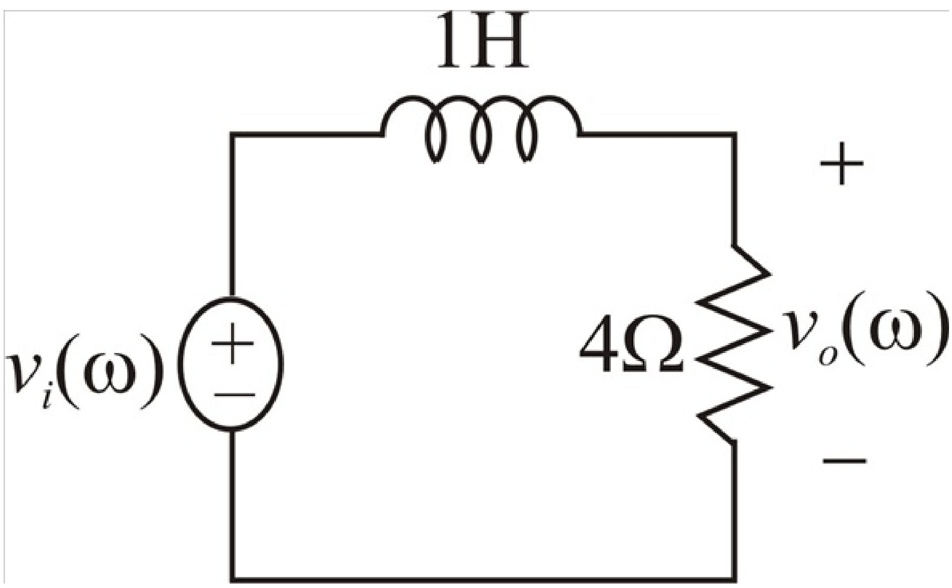
Determine $vo(t)$ in Fig. 18.19 if $vi(t) = 5\text{sgn}(t) = (-5 + 10u(t))$ V.



Step-by-step solution

Step 1 of 10

The given circuit is



Step 2 of 10

The given input voltage is $v_i(t) = 5\text{sgn}(t) = (-5 + 10u(t))$ V

We know that $\mathcal{F}\{1\} = 2\pi\delta(\omega)$

And $\mathcal{F}\{u(t)\} = \pi\delta(\omega) + \frac{1}{j\omega}$

The Fourier transform of the input voltage is

$\mathcal{F}\{v_i(t)\} = \mathcal{F}\{-5 + 10u(t)\}$

Step 3 of 10

$V_i(\omega) = -5\mathrm{F}\{1\} + 10\mathrm{F}\{u(t)\}$

$V_i(\omega) = -5[2\pi\delta(\omega)] + 10\left[\pi\delta(\omega) + \frac{1}{j\omega}\right]$

$V_i(\omega) = -10\pi\delta(\omega) + 10\pi\delta(\omega) + \frac{10}{j\omega}$

$V_i(\omega) = \frac{10}{j\omega}$

Step 4 of 10

In frequency domain,

$R \to R$

$4\ \Omega \to 4\ \Omega$

$L \to j\omega L$

$1\ \mathrm{H} \to j\omega \times 1$

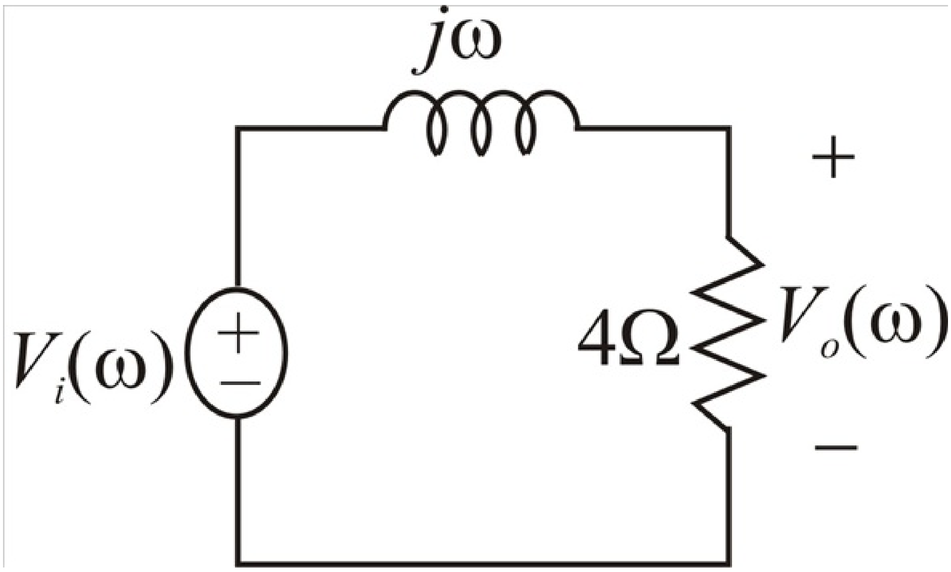
$1\ \mathrm{H} \to j\omega\ \Omega$

$v_i(t) \to V_i(\omega)$

$v_o(t) \to V_o(\omega)$

Step 5 of 10

The circuit is frequency domain is



Step 6 of 10

By voltage division rule,

$V_o(\omega) = \frac{4}{4 + j\omega} V_i(\omega)$

$V_o(\omega) = \frac{4}{4 + j\omega}\left[\frac{10}{j\omega}\right]$

$V_o(\omega) = \frac{40}{j\omega(4 + j\omega)}$

Using partial fractions method

$$\frac{40}{j\omega(4+j\omega)}=\frac{A}{j\omega}+\frac{B}{4+j\omega}$$

$$\frac{40}{j\omega(4+j\omega)}=\frac{A(4+j\omega)+B(j\omega)}{j\omega(4+j\omega)}$$

$$40=A(4+j\omega)+B(j\omega) \quad \text{..........} (1)$$

Consider $j\omega=-4$

$$40=A(4-4)+B(-4)$$

$$40=0-4B$$

$$B=-\frac{40}{4}$$

$$\therefore B=-10$$

Consider constants on both sides of equation (1),

$$4A=40$$

$$\therefore A=10$$

$$V_o(\omega)=\frac{10}{j\omega}-\frac{10}{4+j\omega}$$

We know that

$$\mathrm{F}^{-1}\left\{\frac{2}{j\omega}\right\}=\operatorname{sgn}(t)=-1+2u(t)$$

$$\mathrm{F}^{-1}\left\{\frac{1}{a+j\omega}\right\}=e^{-at}u(t)$$

Using above two equations apply inverse Fourier transform on both sides of $V_o(\omega)$, we get:

$$\mathrm{F}^{-1}\{V_o(\omega)\}=\mathrm{F}^{-1}\left\{5\frac{2}{j\omega}-\frac{10}{4+j\omega}\right\}$$

$$v_o(t)=5\mathrm{F}^{-1}\left\{\frac{2}{j\omega}\right\}-10\mathrm{F}^{-1}\left\{\frac{1}{4+j\omega}\right\}$$

$$v_o(t)=5\left[-1+2u(t)\right]-10\left[e^{-4t}u(t)\right]$$

$$v_o(t)=-5+10u(t)-10e^{-4t}u(t)$$

$$\therefore \boxed{v_o(t)=-5+10\left(1-e^{-4t}\right)u(t)\ \mathrm{V}}$$